RAJALAKSHMI ENGINEERING COLLEGE

V R Krithika Sree 250701371-2025-CSE V2

Started on	Friday, 9 October 2026, 8:49 AM
State	Finished
Completed on	Friday, 9 October 2026, 8:55 AM
Time taken	6 mins 10 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int longest_non_decreasing_subsequence(int arr[], int n) {
    if (n == 0) return 0;

    int dp[n];
    int max_len = 1;


    for (int i = 0; i < n; i++) {
        dp[i] = 1;
    }
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++){
            if (arr[j] <= arr[i] && dp[j] + 1 > dp[i]) {
                dp[i] = dp[j] + 1;
            }
        }
        if (dp[i] > max_len) {
            max_len = dp[i];
        }
    }

    return max_len;
}

int main() {
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("%d\n", longest_non_decreasing_subsequence(a, n));

    return 0;
}

```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course